

Manual

Extended Terminal Functions AIP-EBM 8.2

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Contents

1	Extended terminal functions.....	4
2	Assigning a workplace to several terminals	5
3	Assigning more than 16 workplaces to a terminal	9

1 Extended terminal functions

Purpose

You use the function package if you require one of the following features for the collection of data:

- You want to perform postings for one workplace (e.g. a very large line) in two or three terminals.
- You want to assign more workplaces to one terminal than the maximum number allowed.

Integration

This function can be integrated in the products BDE, MDE, MPL and WRM to extend the possibilities of data collection in the AIP terminal.

Features

- You can assign a HYDRA workplace/machine to more than one terminal (max 3).
- Manual postings are synchronized in all terminals assigned to one machine. The performed postings are promptly displayed on all logically connected devices. Example: An operation has been logged on to machine 24 (terminal 1). The operation at machine 24 should promptly be "visible" on terminal 2.
- You can assign more than 16 workplaces/machines to one terminal. In this context, the data quantity and the performance of the devices is decisive.



Depending on the configuration, specific communication technologies are used that can lead to a different processing logic compared to the processing without this configuration.

We strongly recommend to analyze the requirements in detail and to test the collection and posting processes in a test environment.

2 Assigning a workplace to several terminals

Purpose

You can use the configuration described in this section if you want to collect data as follows:

- A shop floor client (AIP) runs in the so-called combined operation, i.e. the Process Communication Controller (PCC) runs in the same hardware as the AIP and does not have an individual terminal number.
- You want to assign the workplace assigned to this shop floor client (AIP) to one or two further shop floor client(s).

Requirements

In addition to the license, you must make specific configurations. The required configurations depend on the scenario in use.



You cannot combine this terminal configuration with the configuration of a centralized MDE. You can use a shop floor client either for the "combined operation" or the "centralized MDE".

Also note the following:

- If machines are monitored automatically, monitoring may only be performed by exactly one of the terminals (MDE terminal).
- If quantities are recorded automatically, automatic quantities may only be collected by exactly one of the terminals (MDE terminal).
- You must not process data (BDE or MDE processing) in different ways in one terminal. The data of all workplaces assigned to a terminal must be processed in the same way.
- When assigning a workplace to several terminals (maximum 3), the number of terminals should be kept to a minimum. The communication between terminals increases significantly and may have a negative effect on the availability of terminals.
- Workplaces assigned to several terminals must not have a workplace/machine status configured with the control indicator "no order".
- The list of batches logged on to the AIP ("3rd list") is *not* part of the synchronization function.
- The posting volume and the posting frequency in the terminals, especially during a shift change, is important: in a worst-case scenario, there may be problems with synchronization and thus incorrect postings.
- Customizations are *not* compatible by default. We recommend to test this previously.

Configuration: Terminal with MDE data collection

Configure as follows:

MOC: Terminal configuration

Configure the terminal in the [Terminal configuration](#) and set the terminal type to "AIP" (850). Enable the options "Operated as BDE terminal" and "Operated as MDE terminal".

MOC: Workplace terminal assignment

Workplaces for which the terminal records signals: Assign the workplaces to the terminal in the [Workplace terminal assignment](#).

MOC: INI configuration

If the MDE terminal performs machine monitoring plus counter recording, also make the following [INI configurations](#):

Name	MDE
Section	GET_COUNTER_FROM_MDE_TERMINAL
Key	ACTIVE
Value	TRUE
Active	☒
Comment (optional)	

Name	MDE
Section	GET_COUNTER_FROM_MDE_TERMINAL
Key	SHIFT_THRESHOLD
Value	300
Active	☒
Comment (optional)	

The processing logic of these options is described in the document [MBL Distributed MDE Counter.pdf](#).

Configuration ctaip.ini

Activate the direct AIP communication in the *ctaip.ini* of the AIP 8.2.

[DLL]

BusDLL=PCC.EXE

Activate the gateway communication:

[GateWay-Communication]

Active=true

PCC configuration (pcc.ini)

Make the following settings in the local configuration file *pcc.ini* in the PCC directory.

[GateWay-Communication]

```
; Activate the connection "AIP8.2 -> PCC"  
Active=true
```

[HOST]

```
; Activate the connection "PCC -> AIP8.2"  
Active=1
```

[WSK]

```
; Deactivate PDM-connection to HYDRA server  
; Host=<Host name or IP address of the HYDRA server>  
; User=<Terminal number of PCC according to MOC configuration>
```

[Server-Communication]

```
; Deactivate communication to the HYDRA server via EVCOM  
Active=0
```

Terminal restart

Restart the terminal after having configured the AIP.

Configuration: Terminal used for data display and manual input

If you only use the AIP 8.2 for the data display and the manual input, make the following configurations or check these configurations:

MOC: Terminal configuration

In the [Terminal configuration](#), deactivate the option "Operated as MDE terminal" for the terminal.

MOC: Workplace terminal assignment

For all assigned workplaces, you must set the option *Processing* to *Processing according to operation mode of terminal*.

Configuration ctaip.ini

Deactivate the direct AIP communication in the *ctaip.ini* of the AIP 8.2. To do so, comment out the following configuration (precede the rows by a semicolon):

```
; [DLL]  
; BusDLL=PCC.EXE
```

Activate the gateway communication:

[GateWay-Communication]

```
Active=true
```

Configuration hytnrcfg.ini

To enable the transfer/synchronization of the machine status, make the following entry in the hytnrcfg.ini (%HYDRADIR%\<Hydra instance>\custom\aip, e.g. D:\hydra\1\custom\aip\hytnrcfg.ini):

Activate in all terminals:

```
[Tnr Konfiguration 0]
```

```
FollowExternStatus=on
```

Activate only in specific terminals:

You must enter "2000 + terminal number" as number.

Example for terminal 221: $2000 + 221 = 2221$:

```
[Tnr Konfiguration 2221]
```

```
FollowExternStatus=on
```

Terminal restart

Restart the terminal after having configured the AIP.

3 Assigning more than 16 workplaces to a terminal

Purpose

You can assign more than 16 workplaces/machines to one terminal. In this context, the data quantity and the performance of the devices is decisive.

Requirements

Note the following before assigning more than 16 workplaces:

- The number of machines that you can assign to one terminal depends on the posting volume, the data volume and the performance of the terminal hardware.
A big number of postings can result in conflicts during operation. And in case of network failures, a high posting volume results in longer waiting periods until the local queues are emptied.
- The maximum number of machines per terminal is limited to 32 (technical limit).
- The posting volume and the posting frequency in the terminals, especially during a shift change, is important: in a worst-case scenario, there may be problems with synchronization and thus incorrect postings.



We recommend to analyze the requirements in detail and to test the collection and posting processes in a test environment.